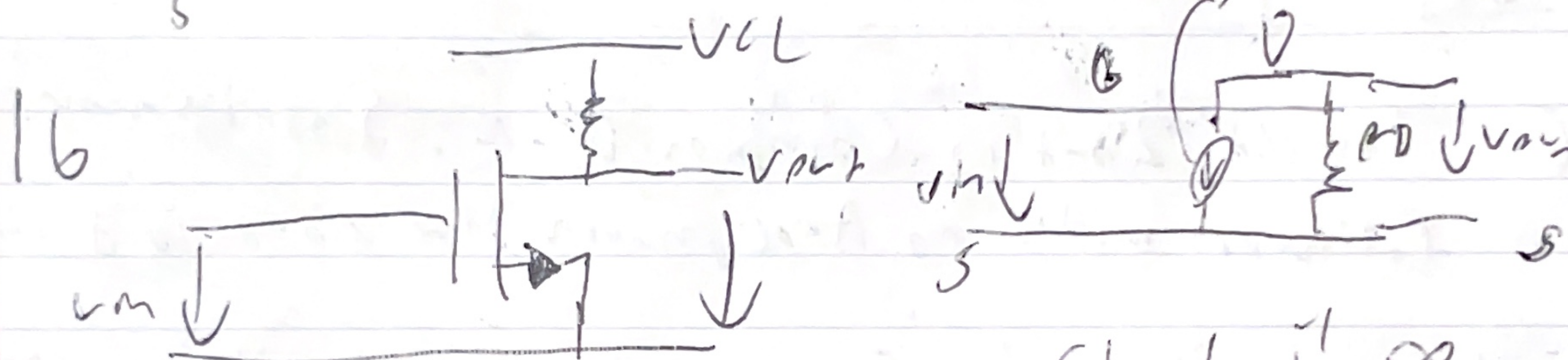
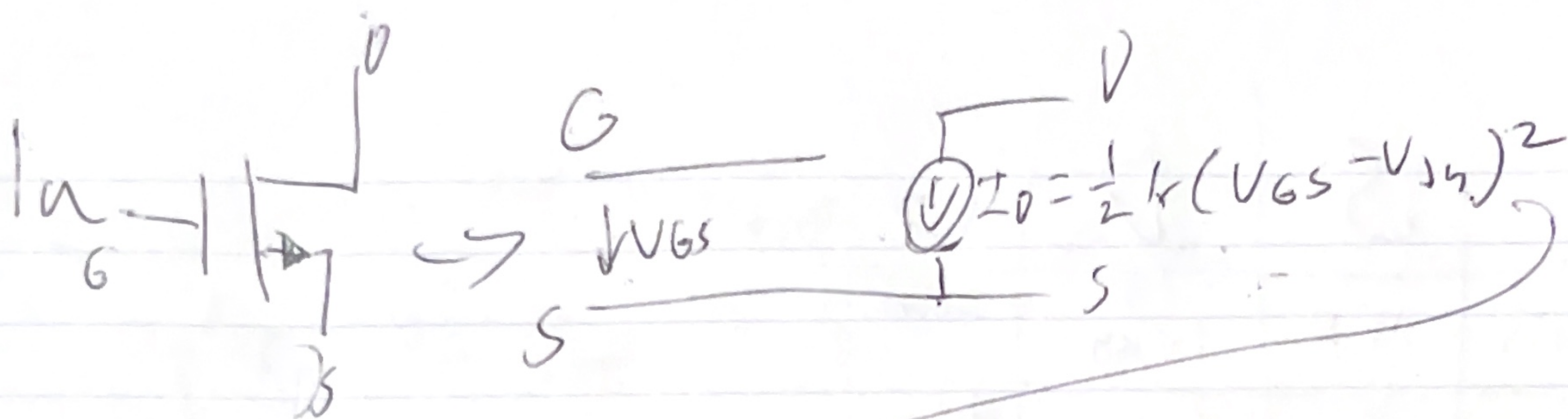
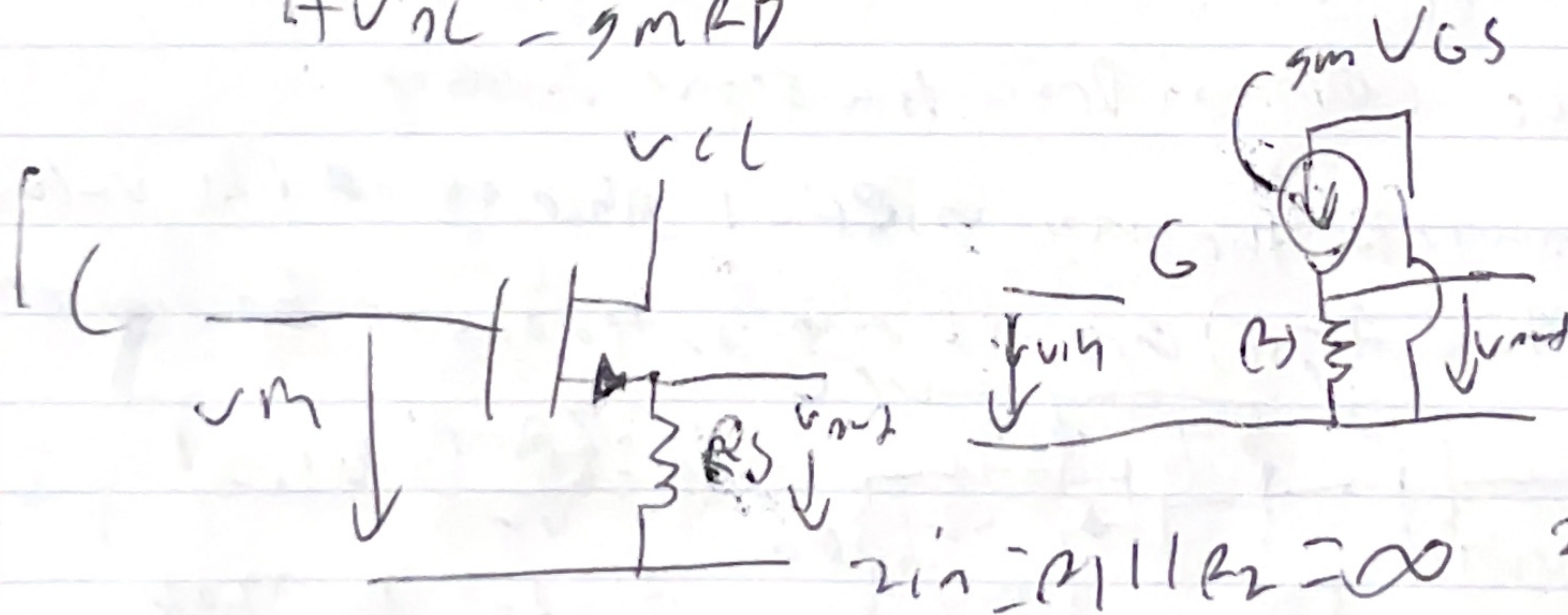




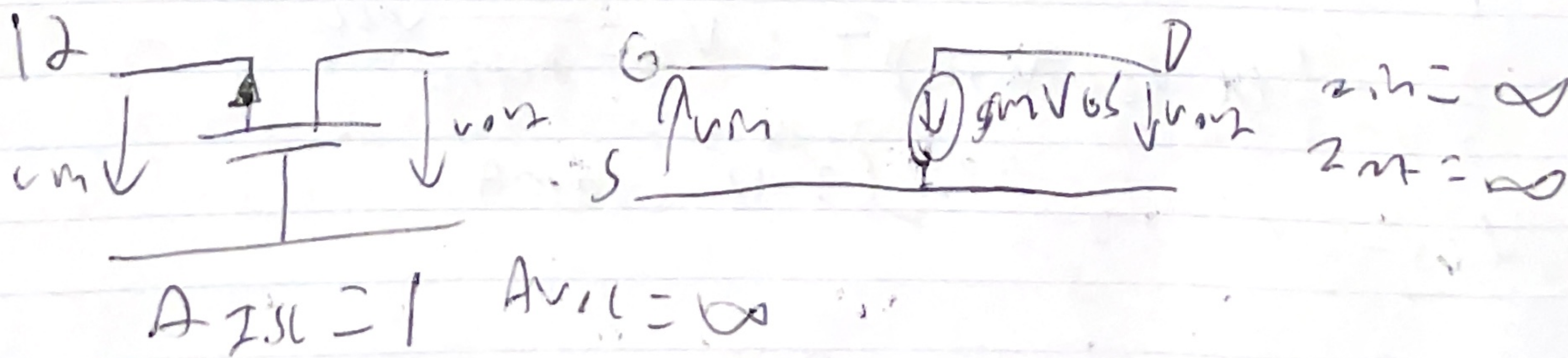
$$z_{in} = R_1 || R_2 \quad R_1, R_2 = \infty \quad \left(\frac{1}{\infty} + \frac{1}{\infty} \right)^{-1} = \frac{\infty}{2} = \infty$$

$$z_{out} = R_D \quad A_{ISC} = g_m (R_1 || R_2) = \infty$$

$$A_{VOL} = g_m R_D$$



$$A_{ISC} = g_m (R_1 || R_2) = \infty \quad A_{VOL} = \frac{g_m R_S}{1 + g_m R_S}$$



$$A_{ISC} = 1 \quad A_{VOL} = \infty$$

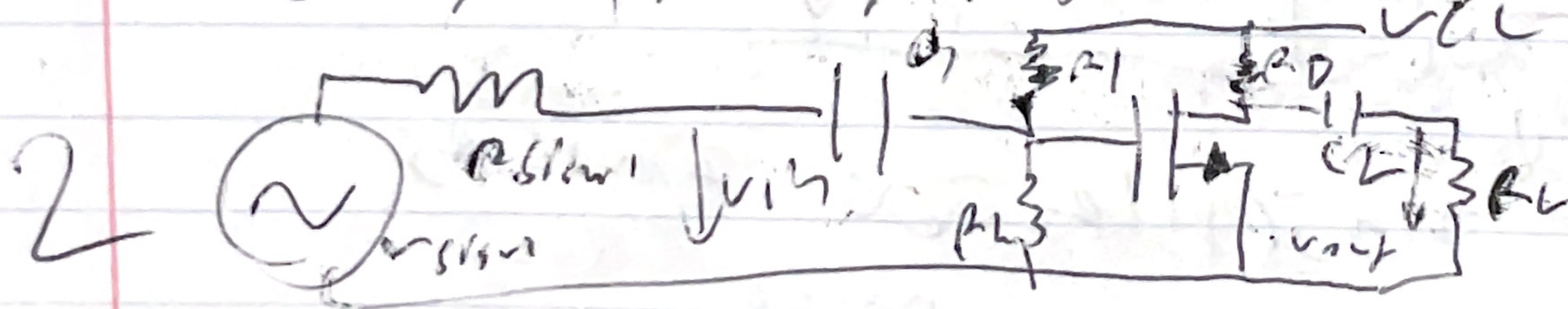
I_e	S	D	G
Z_{in}	mid	high	Low
Z_{out}	mid	Low	High
A_{vol}	mid	Low	High
A_{vsl}	mid	High	Low

1 f Lowest Z_{out} is Common Drain. Since it's a source follower with no A_{vol} , having low Z_{out} is desirable.

1 g Lowest Z_{in} is Common Gate. Since it's a current follower, having low Z_{in} is desirable.

1 h Lowest A_{vol} is Common D which is roughly 1. This is good since Common Drain is a source follower.

1 i Common Source is the most useful since all of its values (Z_{in} , Z_{out} , A_{vol} , A_{vsl}) are mid range, letting it be very versatile.



$$k = \frac{20 \mu A}{V_{th}^2} \quad V_{th} = 1$$

$$R_1 = 80k \quad R_2 = 20k$$

$$R_{sig} = 25k \quad R_L = 1k \quad \text{variable}$$

2 a This is common Source.

$$I_D = \frac{1}{2} k (V_{GS} - V_{th})^2 \quad V_{GS} = \frac{R_2}{R_1 + R_2} V_{CC}$$

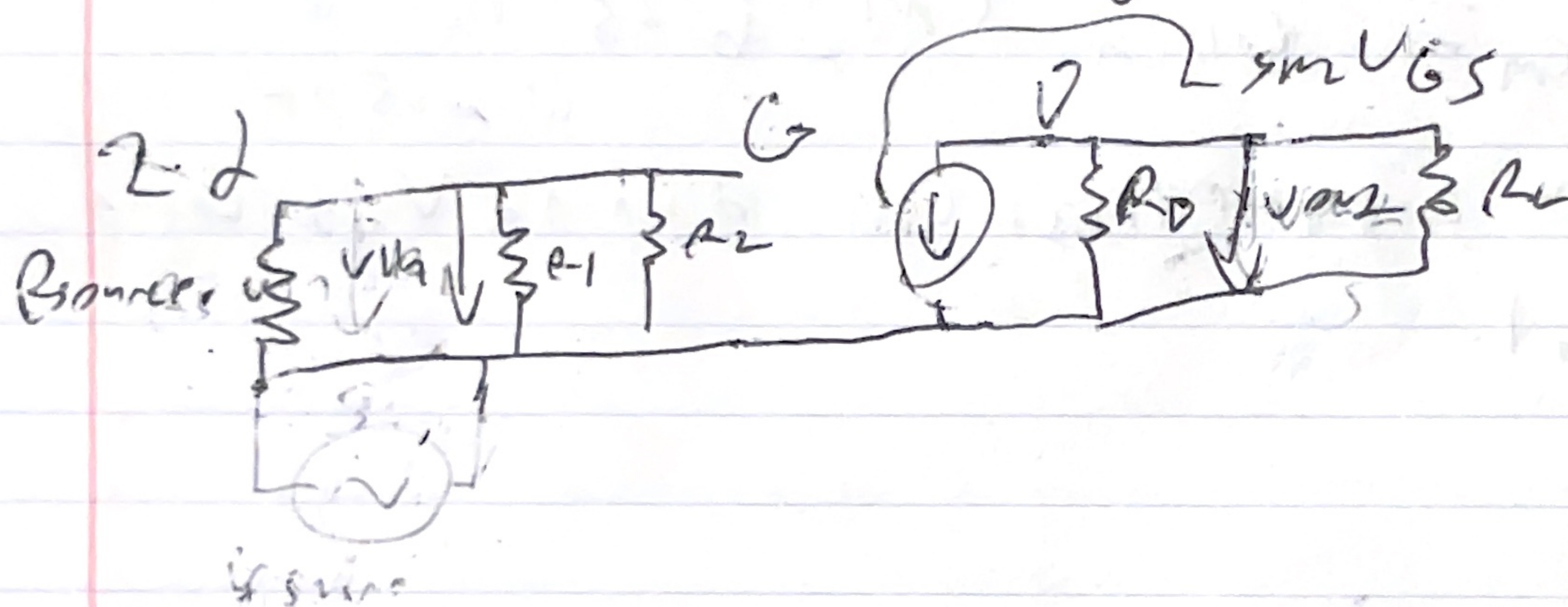
$$V_{GS} = 2V \quad I_D = \frac{20}{2} \cdot \frac{1}{2} (2 - 1)^2 = 10 \mu A$$

21 $V_{DS} = 5$, $I_D = 10 \text{ mA}$

$V_{RD} = I_D R_D$
 $V_{RD} = V_{CC} - V_{DS} - V_{GS}$

$I_D R_D = V_{CC} - V_{DS} \quad 10 \text{ mA } R_D = 5 \quad R_D = 500 \Omega$

22 $g_m = k(V_{GS} - V_{TH}) = \frac{20 \text{ mA}}{V_{GS}}$



23 $Z_M = R_1 \parallel R_2 \parallel R_{source} = 38.19 \Omega$

$Z_{out} = R_D \parallel R_L = 333 \Omega$

$A_{v_{NL}} = \frac{V_{out}}{V_{in}} = \frac{V_{GS} g_m (R_D \parallel R_L)}{V_{GS}} = 20 \cdot 333 = 6.66$

$A_{v_{NL}} = \frac{V_{out}}{V_{in}} = \frac{V_{GS} g_m (R_D \parallel R_L)}{V_{GS}} = 20 \cdot 333 = 6.66$

24 Drain gate capacitance C_{DG} is a parasitic capacitance inside the FET between the drain and gate. It arises due to the fact that capacitors will "exist" between any two sources of voltage that have resistance between them.

Miller capacitance (C_M) is C_{DG} but with different coefficients to fit into the equivalent circuit better.



$$20 \text{ MHz} = \frac{1}{2\pi R_D C_m}$$

$$20 \text{ MHz} = \frac{1}{2\pi R_1 C_m}$$

$$20 \text{ MHz} = \frac{f}{\tau} \quad \tau = 5 \cdot 10^{-8} \text{ seconds} \quad 50 \text{ ns}$$

$$24 \quad 20 \text{ MHz} = \frac{1}{2\pi R_1 C_m} = \frac{1}{618 C_m} \quad C_m = 6.25 \cdot 10^{-13} \text{ F} \\ 0.625 \text{ pF}$$

21 Common Drain has $\approx 1 \text{ pF}$ C_m due to V_{GD} going very close to 1.